

JEE MAIN REVISION MASTER: DEFINITE INTEGRATION 2026 2 QUESTIONS 3.33% WEIGHTAGE ↓33.4%

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 31 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	King's Property & Definite Integral Properties	17 PYQs	Integral Properties
High	Limit of a Sum & Area Under Curves	14 PYQs	Summation & Area

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
King's Property & Definite Integral Properties	$\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$, $\int_{-a}^a f(x) dx = 0$ (if odd), $\frac{d}{dx} \int u(x)v(x) f(t) dt = f(v(x))v'(x) - f(u(x))u'(x)$
Limit of a Sum & Area Under Curves	$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n f\left(\frac{r}{n}\right) = \int_0^1 f(x) dx$, $\text{Area between } y^2 = 4ax \text{ and } y = mx \text{ implies } \frac{8a^2}{3} m^3$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 King's Property & Definite Integral Properties (Very High Yield — 17 PYQs)

Core Principle: King's property $f(a+b-x)$, reflection symmetry, integrals of even/odd functions, periodic functions, and Leibnitz integral differentiation.

■ **Caution / JEE Pitfall:** King's property added to the original integral often yields $2I = \int_a^b 1 dx = b - a \implies I = (b - a) / 2$.

■ **Speed Trick:** Integral of odd function over symmetric limits $[-a, a]$ is always ZERO: $\int_{-a}^a f(x) dx = 0$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $\int_{-6}^6 (a-6)^b - 6f(x+6) dx$ is equal to..." [Ans: C]
- [2026 Shift] Q#2 (MCQ): "One of the possible functions $f(x)$ which satisfies $\int_{-2}^2 f(x) dx = 0$ is..." [Ans: A]
- [2025 Shift] Q#4 (MCQ): "The value of $\int_0^{2\pi} \sqrt{1 + \sin((x/2))} dx$ is..." [Ans: N/A]

3.2 Limit of a Sum & Area Under Curves (High Yield — 14 PYQs)

Core Principle: Definite integrals as limits of Riemann sums ($\lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \cdot \Delta x$), and enclosed area calculations between curves.

■ **Caution / JEE Pitfall:** Area is always a positive geometric quantity; if the curve dips below the x-axis, take absolute value $|\int y, dx|$.

■ **Speed Trick:** Area bounded by $y^2 = 4ax$ and $x^2 = 4by$ is given directly by $\frac{16ab}{3}$.

Representative Source PYQs:

- [2025 Shift] Q#1 (MCQ): " $\int_0^1 \log\left(\frac{1}{x}-1\right) dx$ is..." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "The value of $\int_{-1}^1 \sin^5 x \cos^4 x dx$ is..." [Ans: N/A]
- [2024 Shift] Q#3 (MCQ): " $\int_1^2 (1+5(|x-3|+|1-x|)) dx = \dots$ " [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$\int_{a-6}^{b-6} (a-6)^b - 6f(x+6) dx$ is equal to

- (A) $\int_{a-6}^{b-6} (a-6)^b f(x+6) dx$
- (B) $\int_{a-6}^{b-6} (a-6)^b f(x-6) dx$
- (C) $\int_{a-6}^{b-6} (a-6)^b f(x) dx$ [Correct]
- (D) $-\int_{a-6}^{b-6} (a-6)^b f(x) dx$

Correct Answer: C

Solution Analysis: Put $t=x+6$ Then $dt=dx$ Now change the limits. When $x=a-6$, we get $t=(a-6)+6=a$ When $x=b-6$, we get $t=(b-6)+6=b$ So, $\int_{a-6}^{b-6} (a-6)^b - 6f(x+6) dx = \int_a^b (a-6)^b - 6f(t) dt$ The letter used for the variable does not matter in a definite integral. So, $\int_a^b (a-6)^b f(t) dt = \int_a^b (a-6)^b f(x) dx$ Hence, $\int_{a-6}^{b-6} (a-6)^b - 6f(x+6) dx = \int_a^b (a-6)^b f(x) dx$ So the correct option is Option C

Benchmark Problem 2: KCET 2025 (Q#1)

$\int_0^1 \log\left(\frac{1}{x}-1\right) dx$ is

- (A) 1
- (B) 0
- (C) $\log_e 2$
- (D) $\log_e\left(\frac{1}{2}\right)$

Correct Answer: None

Solution Analysis: $\int_0^1 \log\left(\frac{1}{x}-1\right) dx = \int_0^1 \log\left(\frac{1-x}{x}\right) dx = \int_0^1 \log(1-x) dx - \int_0^1 \log(x) dx$
 $I = \int_0^1 \log(1-x) dx - \int_0^1 \log(x) dx$
 $I = \int_0^1 \log(1-x) dx - \int_0^1 \log(1-x) dx = 0$